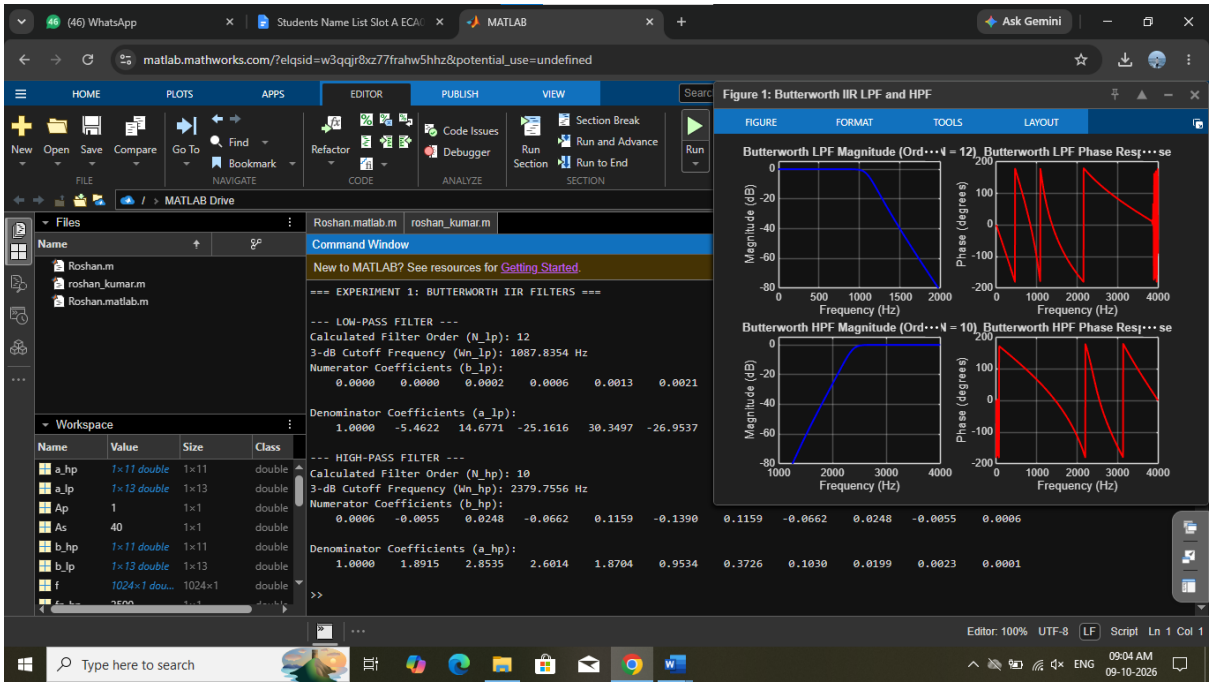

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MATLAB
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Refactor
Code Issues
Debugger
Run Section
Section Break
Run and Advance
Run to End
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NAVIGATE
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SECTION
MATLAB Drive
Files
Name
Roshan.m
roshan_kumar.m
Roshan.matlab.m
Workspace
Name
Value
Size
Class
a_hp
1×11 double
1×11
double
a_lp
1×13 double
1×13
double
Ap
1
1×1
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As
40
1×1
double
b_hp
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double
b_lp
1×13 double
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Roshan.matlab.m
roshan_kumar.m
Command Window
New to MATLAB? See resources for Getting Started.
=== EXPERIMENT 1: BUTTERWORTH IIR FILTERS ===
--- LOW-PASS FILTER ---
Calculated Filter Order (N_lp): 12
3-dB Cutoff Frequency (Wn_lp): 1087.8354 Hz
Numerator Coefficients (b_lp):
0.0000 0.0000 0.0002 0.0006 0.0013 0.0021
Denominator Coefficients (a_lp):
1.0000 -5.4622 14.6771 -25.1616 30.3497 -26.9537
--- HIGH-PASS FILTER ---
Calculated Filter Order (N_hp): 10
3-dB Cutoff Frequency (Wn_hp): 2379.7556 Hz
Numerator Coefficients (b_hp):
0.0006 -0.0055 0.0248 -0.0662 0.1159 -0.1390 0.1159 -0.0662 0.0248 -0.0055 0.0006
Denominator Coefficients (a_hp):
1.0000 1.8915 2.8535 2.6014 1.8704 0.9534 0.3726 0.1030 0.0199 0.0023 0.0001
>>
Figure 1: Butterworth IIR LPF and HPF
FIGURE
FORMAT
TOOLS
LAYOUT
Butterworth LPF Magnitude (Ord···N = 12)
Butterworth LPF Phase Resp···se
Magnitude (dB)
Phase (degrees)
Frequency (Hz)
Butterworth HPF Magnitude (Ord···N = 10)
Butterworth HPF Phase Resp···se
Editor: 100%
UTF-8
LF
Script
Ln 1 Col 1
Type here to search
ENG
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09-10-2026